Relay Controller - Serial Commands and Arduino Pro Mini Pinout

Serial Commands

All commands can be sent to `Serial` at **9600 baud**. Commands are case-insensitive.

Command List

Command	Arguments	Description
`set-timeout {color} {seconds}`	Color name, Timeout in seconds	Set relay auto-off timeout.
`set-debounce {color} {milliseconds}`	Color name, Debounce time in ms	Set button debounce time.
`on {color}`	Color name	Turn a relay ON manually.
`off {color}`	Color name	Turn a relay OFF manually.
`save`	None	Save current timeout and debounce settings to EEPROM.
`read-timeout {color}`	Color name	Read current timeout setting for a relay.
`read-debounce {color}`	Color name	Read current debounce setting for a relay.
`status {color}`	Color name	Get current latch status and timer ticks.

Example Usage

```
set-timeout blue 600   // Sets blue relay timeout to 10 minutes (600 seconds)
set-debounce red 250   // Sets red button debounce to 250 ms
on green               // Turns green relay ON
off yellow             // Turns yellow relay OFF
save                   // Saves current timeout and debounce settings
read-timeout blue      // Reads blue relay timeout
status red             // Shows red relay status
```

Arduino Pro Mini Pinout (Relay Connections)

Signal	Arduino Pin	Description

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Blue Button	15 (A1)	Input button for blue relay
Green Button	14 (A0)	Input button for green relay
Yellow Button	16 (A2)	Input button for yellow relay
Red Button	10 (D10)	Input button for red relay
Blue LED	A3	Status LED for blue relay
Green LED	A2	Status LED for green relay
Yellow LED	A1	Status LED for yellow relay
Red LED	A0	Status LED for red relay
Blue Relay Control	6 (D6)	Output to control blue relay
Green Relay Control	7 (D7)	Output to control green relay
Yellow Relay Control	8 (D8)	Output to control yellow relay
Red Relay Control	9 (D9)	Output to control red relay

Notes:

- All button pins are set to `INPUT_PULLUP` mode.
- Relays are **active LOW** (0 = ON, 1 = OFF).
- LEDs are **active HIGH**.

Default Settings

Setting	Value
Timeout	30 minutes (1800 seconds)
Debounce	200 milliseconds

EEPROM memory is automatically initialized with default values if invalid (0 or 0xFFFFFFFF) data is found.